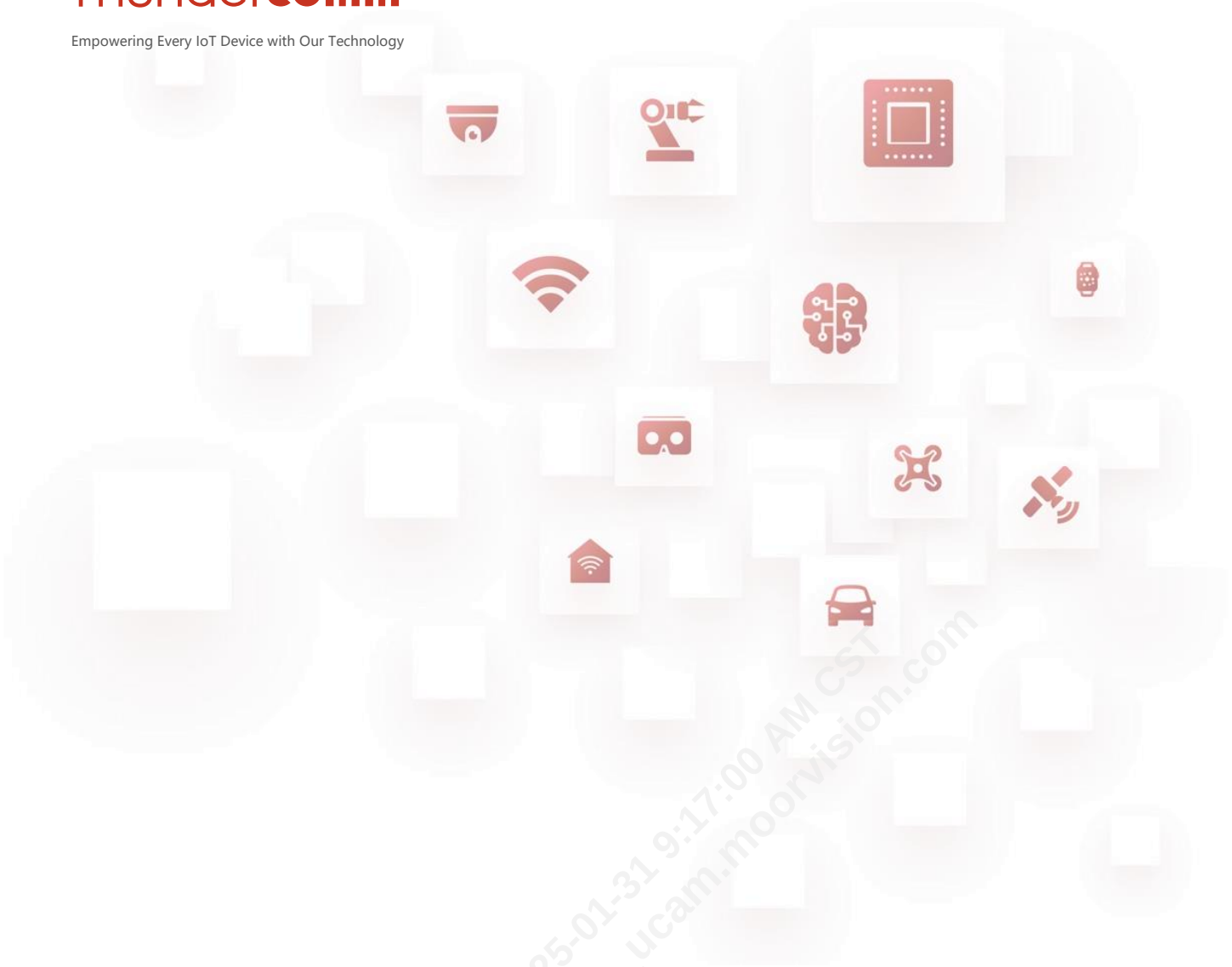
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Thundercomm TurboX™ C410/C610 Open Kit Linux Enablement

Software Release Notes

Rev. V1.2

July 06, 2021

DN: tc-gle-2412

SDK Version: FC1.r001003

Revision History

Revision	Date	Description
1.0	Sep 21, 2020	First official release
1.1	Oct 9, 2020	Update "Table 4-1 LE SDK Information": Change the "Baseline Distribution" from "CS1 0.0.070.1" to "CS2 0.0.071.1" Update "4 Getting Started with LE SDK": Change "<user>" to "<User Email>" in repo init command. Update "4.6 Flash Images to Board" Add the NOTE at the beginning of this chapter. Delete -a in turbox_flash.sh command.
1.2	July 06, 2021	Update the SDK code download command in 4.4.2.Download the LE SDK code.

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Glossary

ADB	Android Debug Bridge
GCC	GNU Compiler Collection
GNU	GNU is not Unix
GUI	Graphic User Interface
LE	Linux Enablement
LTS	Long Term Support
SSH	Secured Shell
USB	Universal Serial Bus
LDC	Lens Distortion Correction

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1 Introduction

This document provides information on how to obtain, build, and program QCS610 Linux Enablement (LE) SDK into the TurboX™ C610 Platform.

The Linux Embedded Software Development Kit supports the Linux operating system on the Thundercomm TurboX™ SOM. The goal of the LE SDK is to provide a simple and flexible development platform to rapidly build IoT products with the QTI chipsets. The LE SDK provides the build framework, software components, tools, and sample applications to evaluate the platform and build software images for the target product.

The LE SDK is based on the Yocto Project 2.6 release and follows several of its development principles. The Yocto Project is used by the LE SDK for the build framework.

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2 System Requirements

Table 2-1 lists the hardware and software required to install and run the LE SDK.

Table 2-1 System Requirements

Hardware / Software Requirements	Source / Vendor	Function
Standalone hardware system minimum requirement: ▪ 8 GB RAM ▪ Quad core CPU **NOTE:** System with lower hardware specifications will increase compilation time. About an hour is the compilation time for a system with 16 GB RAM and Intel i7-2600 @3.4 GHz.	–	SDK build machine
Ubuntu 16.04 LTS Linux distribution for 64 bit architecture	Ubuntu Community/ Canonical, Ltd.	SDK build host OS
Repository	SDK Open Source Project	SDK source management tool
Python 2.7.12 with sqlite3 module	Python.org	Building subsystem

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3 Set Up the Development Environment

This chapter describes how to set up the Linux development environment on an Ubuntu host machine.

Prerequisites:

The build host must meet the following requirements:

- Host machine Linux distro – 16.04 LTS
- Minimum 90 GB disk space
- tar v1.27 or newer
- Privileges to install utilities
- Install the host packages

3.1 Ubuntu 16.04 (64-bit) Installation

(1) **Install ubuntu 16.04 on the computer following the instructions at:**
http://www.ubuntu.com/download/desktop/install-ubuntu-desktop

(2) **Perform a software update using one of the following options:**
In the GUI, select *System>Administration >Update Manager*.

3.2 ADB, Fastboot Tools and USB Drivers Installation

This section provides instructions for installing ADB(v1.0.36 or later) and Fastboot tools, and USB drivers on the host machine.

The Fastboot tool communicates with the bootloader. This tool flashes the images onto the board.

The ADB tool communicates with the device and provides access to the Unix shell allowing to push, modify, debug, and tweak system files.

3.2.1 ADB and Fastboot Tools Installation

To install ADB and Fastboot tools, execute the following commands:

```
$ sudo apt-get install android-tools-adb
$ sudo apt-get install android-tools-fastboot
```

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3.2.2 USB Drivers – Installation and Configuration

To install and configure USB drivers, follow the below sequence:

(1) **Setup the udev rules on the Linux PC:**

a. Login as *root* or *sudo* and navigate to the directory /etc/udev/rules.d/

b. Add the following rules in file 99-android.rules:

```
SUBSYSTEM=="usb", ATTR{idVendor}=="18d1", MODE="0777", GROUP="adm"
SUBSYSTEM=="usb", ATTR{idVendor}=="05c6", MODE="0777", GROUP="adm"
```

c. Restart the udev service by executing the commands:

```
$ sudo chmod a+r /etc/udev/rules.d/99-android.rules
$ sudo service udev restart
```

(2) **Verify that the installation was successful:**

Connect the board to the PC with USB cable and power on the target board to start the system. The shell will display:

```
$ adb devices
List of devices attached
7050372c            device
```

3.3 Required Build Environment

3.3.1 GCC/G++

NOTE: Use Ubuntu 16.04 along with GCC/G++ v5.4.

(1) **Check version**

Run the following commands to ensure GCC/G++ version 5.4 is installed:

```
$ gcc --version
gcc (Ubuntu 5.4.0-6ubuntu1~16.04.12) 5.4.0 20160609
Copyright (C) 2015 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
$ g++ --version
g++ (Ubuntu 5.4.0-6ubuntu1~16.04.12) 5.4.0 20160609
```

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```
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```

(2) **Update version**

If the host machine has an older version, complete the following steps to upgrade to version 5.4:

a. Run the following commands:

```
$ sudo add-apt-repository ppa:ubuntu-toolchain-r/test
$ sudo apt-get update
$ sudo apt-get install gcc-5-multilib g++-5-multilib
$ sudo update-alternatives --install /usr/bin/gcc gcc /usr/bin/gcc-5  100
$ sudo update-alternatives --install /usr/bin/g++ g++ /usr/bin/g++-5  100
```

b. On the shown prompt Select gcc-5 and run the following command:

```
$ sudo update-alternatives --config gcc
```

c. Select g++-5 again and run the following command:

```
$ sudo update-alternatives --config g++
```

3.3.2 Linux Package Dependencies

(1) **Dependency Library is required to be installed on the host PC.**

To install the library, run the below cmd

```
$ sudo apt-get install gawk wget git-core diffstat unzip texinfo gcc-multilib \
build-essential chrpath socat cpio python python3 python3-pip python3-pexpect \
xz-utils debianutils iputils-ping libsdl1.2-dev xterm openjdk-8-jdk
```

To install the XML::Simple of perl, run the below cmd

```
$ sudo cpan install XML::Simple
```

(2) **Check that the bash shell is in use**

```
$ sudo rm /bin/sh
$ sudo ln -sf /bin/bash /bin/sh
```

(3) **Build dependencies**

Prior running the build command , the following packages must be installed: diffstat, makeinfo, and chrpath.

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For build environment dependencies, go to the Yocto Project Active Release Documentation page at https://www.yoctoproject.org/docs/2.6.2/brief-yoctoprojectqs/brief-yoctoprojectqs.html

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4 Getting Started with LE SDK

4.1 LE SDK Information

Table 4-1 presents the LE SDK information.

Table 4-1 LE SDK Information

Product	Baseline Distribution	SDK Version
QCS610.LE.1.0	CS2 0.0.071.1	FC1.r001003

4.2 LE SDK Download

The next steps explain how to set an account to download the SDK.

4.2.1 Account Activation

Add the account/password information to $HOME/.netrc by running the following command.

```
$ echo "machine partner.thundercomm.com login <User Email> password <password>" >> ~/.netrc
```

➲NOTE: Please use the email and password you have registered at www.thundercomm.com to replace <User Email> and <password> in the above command.

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4.2.2 Download the LE SDK Code

To download the LE SDK run the following commands:

1. Initialize the repository

China area:

```
$ repo init -u https://partner.thundercomm.com/sdkgit/a/manifest -b SDK.turbox-c610-le1.0-dev -m turbox-c610-le1.0-dev.fc1.r001003_cn.xml --repo-url=https://partner.thundercomm.com/sdkgit/a/tools/repo --repo-branch=stable --no-repo-verify
```

NOTE: The opensource part is downloaded from sdk gerrit of TC.

Out of china areas:

```
$ repo init -u https://partner.thundercomm.com/sdkgit/a/manifest -b SDK.turbox-c610-le1.0-dev -m turbox-c610-le1.0-dev.fc1.r001003.xml --repo-url=https://partner.thundercomm.com/sdkgit/a/tools/repo --repo-branch=stable --no-repo-verify
```

NOTE: The opensource part is downloaded from codeaurora.

2. Sync the source code

```
$ repo sync -c -j4 --no-tags
```

NOTE: Please use the email you registered on Thundercomm website to replace <User Email> in the above commands, for example:
repo init -u ssh://service@thundercomm.com@106.2.164.79:9418/...

4.3 Fast download ways based on old SDK

If you have downloaded the old SDK, there are two ways to download the new SDK fast.

1. Upgrade the local old folder to new release

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Going to the old local code folder, you can run the repo downloading command of new release as 4.2.2 chapter, and then run the "repo sync -j4 -c --no-tags" command to receive the new code.

2. Download the new release to new a folder

Firstly, creat the new folder, and run the repo init command with --reference parameter. For example:

```
$ repo init -u ssh://<User Email>@106.2.164.79:9418/manifest -b SDK.turbox-c610-le1.0-dev -m turbox- \
c610-le1.0-dev.fc1.r001003.xml --reference=<old_sdk _code>
$ repo sync -j4 -c --no-tags
```

NOTE: --reference=<old_sdk_code path> and the old_sdk_code path is the absolute path of your old sdk path, For example: --reference=/home/host/turbox-c610-le1.0-dev.es.r000002

4.4 Build Image

To compile AP code, run the following script file in the source code root directory:

Build debug version:

```
$ ./turbox_build.sh -a -l -v debug
```

Build perf version:

```
$ ./turbox_build.sh -a -l -v perf
```

Build user version:

```
$ ./turbox_build.sh -a -l -v user
```

NOTE: Uart debug can not be used in perf/user build.

4.5 ROM package

(1) **Run the following command to update common info:**

Debug version:

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```
$ ./turbox_build.sh -u --uci-imf -v debug
```

Perf version:

```
$ ./turbox_build.sh -u --uci-imf -v perf
```

User version:

```
$ ./turbox_build.sh -u --uci-imf -v user
```

Finally, the following log will be printed, indicating that the command was successfully executed

```
- update_common_info.py:======  UPDATE COMMON INFO COMPLETE=======
```

(2) **Then run the following command to pack the ROM package.**

```
$ ./turbox_build.sh --zip_fastboot_build -l -v perf
```

The ROM package will be placed in the turbox/output directory.

(3) **Get the ROM package provided by Thundercomm**

Customers can get ROM package provided by Thundercomm from the following URL:

FastbootBuild_Turbox-C610_le1.0.r.debug.fc1.r001003.zip
FastbootBuild_Turbox-C610_le1.0.r.perf.fc1.r001003.zip
FlatBuild_Turbox-C610_le1.0.r.debug.fc1.r001003.zip
FlatBuild_Turbox-C610_le1.0.r.perf.fc1.r001003.zip

4.6 Flash Images to Board

NOTE: Please flash the FLatBuild Image to the Open Kit firstly if the current Image is not this release on your Open Kit.

4.6.1 Flash FastbootBuild image

Fastboot tool flash image, the device must be placed in the Fastboot mode. Use the following methods:

When the device is in the ADB mode, enter the following command:

```
$ adb root
$ adb reboot bootloader
```

To determine if the device has entered the Fastboot mode, run the following command:

```
$ fastboot devices
```

If successful, an output like the following displays on the shell:

```
7050372c fastboot
```

Execute the below commands to flash an image to the board:

```
$ cd FastbootBuild_Turbox-C610_le1.0.r.perf.fc1.r001003
```

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```
$bash ./turbox_flash.sh -r -v perf
```

NOTE: ./turbox_flash -h will show detailed usage!

```
./turbox_flash -h
Usage:
  bash ./turbox_flash.sh [options] [options]
  The images except gpt,userdata,persist will be flashed if no options input
Description:
  flash images with fasboot
Options:
  -h, --help     display this help message
  -v, --variant <debug/perf/user>       set the build variant.(default: debug)
  -f, --force    gpt,persist and userdata will be flashed.
                 User data and calibration data(BT/WIFI MAC address, etc..) will be erased.
  -r, --reboot          reboot the device automatically after flashing completed.
  -b, --nonhlos         flash all bp images(xbl,rpm,ftfm,tz,dsp,modem,pmic).
  -a, --hlos            flash all ap images(aboot,bootimage,recovery,system,vendor,cache,mdtp).
  --all             flash all images
Example:
  ./turbox_flash.sh
       Flash the images except gpt,userdata,persist.
  ./turbox_flash.sh --force
       Flash all images.
  ./turbox_flash.sh --hlos
       Flash all ap images.
```

4.6.2 Flash FaltBuild Image

If the Open kit can't bootup to fastboot mode, you can flash the FlatBuild image using QDL.

1. Enter 9008 mode

Press and hold the Force_USB_Boot key , then insert the usb and power cable, and you can see the 9008 port, and release the Force USB Boot key:

```
$ lsusb
......
Bus 001 Device 042: ID 05c6:9008 Qualcomm, Inc. Gobi Wireless Modem (QDL mode)
.......
```

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2. Start to flash the flatbuild image

   The download command is as below:

```
$ cd FlatBuild_Turbox-C610_le1.0.r.perf.fc1.r001003
$ chmod +x qdl
$ sudo ./qdl --storage emmc prog_firehose_ddr.elf    rawprogram*.xml patch*.xml
```

5 New Features in this release

- Add OpenCV-3.4.10
- Add Multiplex Decoding Demo app
- Add Face Detection Demo app
- Enable Multi-Streaming
- Enable Digital Zooming
- Add gtk+3_3.18.8
- Support new hardware version

6 Know Issues in this release

- Camera LDC does not work well.
- System can not enter to sleep mode.

7 Feature List(to be update)

Feature	status

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8 Test Report(to be update)

Test Case	status

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